

CFD Assessment of a NACA 4412 Airfoil for Urban Vertical Axis Wind Turbines

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1. Introduction

Wind energy is a key renewable resource for reducing reliance on fossil fuels and decarbonising electricity generation. Large horizontal axis wind turbines (HAWTs) dominate utility scale applications, vertical axis wind turbines (VAWTs), however, offer potential advantages in built up environments due to key features such as omnidirectional operation, don't require yaw mechanisms, simpler structural layouts and are more easily integrated at ground level or in buildings (Bošnjaković et al., 2025). However, the aerodynamic performance of VAWTs is highly sensitive to the airfoil and blade cross-section design, typically showing lower power coefficients than HAWTs, often by ~25% under similar conditions, making the choice and optimisation of the airfoil a critical factor in determining energy capture, especially under the complex flow conditions found in cities (Bošnjaković et al., 2025; Zakis et al., 2026).

The lift and drag characteristics of airfoil sections, along with their stall behaviour, directly influence the torque and power that can be extracted across a range of operating conditions. In this study Computational Fluid Dynamics (CFD) is employed to simulate the flow around an airfoil and to quantify aerodynamic coefficients, providing a detailed understanding of how the profile performs at representative urban wind speeds.

Urban wind environments are characterised by highly turbulent, unsteady and non-uniform flows due to the presence of buildings and other obstacles, which can lead to lower average wind speeds and significant fluctuations in its magnitude and direction (Micallef & Van Bussel, 2018). For example, measurements at one urban site showed turbulence intensity of ~24 vs ~18% in IEC open terrain models, emphasizing how urban conditions can exceed standard design assumptions (Zhang et al., 2023). For VAWTs operating in these conditions the effective Reynolds number and angle of attack experienced by the blades vary continuously, so airfoils must maintain acceptable lift to drag performance over a wide range of conditions rather than only near a single design point (Micallef & Van Bussel, 2018). In this context, improving aerodynamic robustness of the blade section at both low and higher wind speeds is essential to improve the overall efficiency and reliability of small scale wind energy systems.

The aim of this study is to analyse and modify the aerodynamic performance of a NACA4412 airfoil, for integration into a VAWT under typical urban conditions, using CFD simulations at two distinct wind speeds. The specific objectives are

- To simulate the flow around the original airfoil at wind speeds of 3 and 15 m/s at fixed angle of attack (AOA) of 10°.
- Perform a mesh independence study at 15 m/s to ensure the predicted aerodynamic coefficients are independent of mesh coarseness.
- To design and implement a clear geometric modification of the NACA 4412 airfoil, such as a added feature or shape alteration, intended to better the performance in a VAWT context.
- And to evaluate and compare the lift, drag and flow structures of the original and modified airfoils at both wind speeds, drawing conclusions on their suitability for urban wind energy applications.

2. Methodology

2.1. Airfoil Selection

The baseline section analysed in this work is the NACA 4412 airfoil with chord length c=0.4m. The NACA digits indicate a maximum camber of 4% located at 40% of the chord from the leading edge and a maximum thickness of 12% of the chord. This cambered profile is therefore expected to generate meaningful lift at moderate angles of attack, which is relevant for VAWT blades that must produce aerodynamic torque under varying inflow conditions.

For the case in this study the Reynolds number (chord based , $Re_c = \frac{U_\infty c}{\nu}$), spans ~8.5x10^4 at 3m/s to ~4.3x10^5 at 15m/s (computed with same constants used in fluent), a range where transition related phenomena (including laminar separation bubbles) can occur and affect lift, drag and separation behaviour (Golmirzaee & Wood, 2024). This makes NACA 4412 a useful baseline because it is used in the wind energy context and has published datasets that can be used as sanity checks on whether CFD trends look physically reasonable, even though many classic datasets are at higher Reynolds numbers than the present case.

In the context of VAWTs, the blade section's lift and drag directly influence the torque, so quantifying baseline C_l, C_d and flow features at the two specified wind speeds provides a good reference for judging whether the modification improves performance under urban conditions.

2.2. Modification Techniques

To improve VAWT blade section efficiency under urban wind conditions, the baseline NACA 4412 was modified using suction side dimples intended to influence near wall mixing and separation behaviour at low to moderate Reynolds numbers. As a simple parametric comparison, two configurations were tested, a singular semicircular indent and a two dimple variant in a 2D model, with placement and size selected directly from the methodology in (Xie et al., 2022), who follow recess-design guidance showing that the most effective location is just upstream of the laminar separation bubble (LSB) and that the depth/diameter ratio should lie in an optimum band. Therefore, the dimple centres were set at x/c = 0.35 and 0.5 (x=0.14, 0.2 m), with depth h=0.01c (4mm) and printed diameter/width d = 0.08c (32mm) giving h/d = 0.125 within the recommended 0.1-0.15 range cited in the study.

A key limitation however, is that separation location and boundary layer thickness vary with Reynolds number and AoA, the reference notes that at low enough Re the LSB may cover the dimple and reduce its effectiveness, so this should be treated as a literature design rather than an optimised configuration which would require a broader parametric study.

2.3. CFD Setup

2.3.1. Geometry Creation

The computational geometry was created in ANSYS ICEM CFD as a two dimensional C type domain surrounding the NACA 4412 airfoil following common best practice for modelling sectional airfoil flow as 2D for surface pressure and aerodynamic coefficients (Golmirzaee &

Wood, 2024). The domain dimensions were set to 20 chord lengths (c=0.4m) upstream of the airfoil's aerodynamic centre (~1/4 from the leading edge), 30 chord lengths downstream and 20 chord lengths perpendicular to the freestream direction on both sides, resulting in a blockage ratio of below 3% (Bashir et al., 2024). This configuration minimises artificial blockage effects and makes sure that boundary conditions don't influence the flow field around the airfoil, as C-meshes with boundaries less than 30c are common practice for external aerodynamic simulations but require domain independence studies to limit errors (Golmirzaee & Wood, 2024).

2.3.2. Mesh Generation

A structured blocking topology was employed in ICEM CFD featuring a high resolution O-grid (split to C-grid configuration) encircling the airfoil for orthogonal cells for accurate boundary layer resolution around high curvature areas like the leading edge (Bashir et al., 2024). Interfaced structured blocks (H-type via downstream splits) for the near field, then converted to an unstructured quadrilateral dominant mesh format across the whole domain to enable Fluent compatibility while retaining the structured multiblock topology.

Near wall resolution was achieved through structured edge parameters on airfoil surfaces, with first cell spacing of 1×10^{-5} m and approximately 70 nodes per edge with a growth ratio of 1.2. To obtain a priori estimate of first cell height targeting y^+ of < 1 to correctly resolve the laminar and transitional boundary layers and adhere to turbulence model requirements (Bashir et al., 2024; Golmirzaee & Wood, 2024), the first cell location should satisfy equations shown in (László Könözsy, 2026):

$$y_1 \leq \frac{y^+ \nu}{u_\tau}, \qquad u_\tau = U_e \sqrt{\frac{C_f}{2}}, \qquad Flat\ plate: \frac{\overline{C_f}}{2} \approx 0.037\, Re_L^{-\frac{1}{5}}$$

Using standard constants for air and assuming the airfoil as a flat plate an estimated first cell distance of 1.8×10^{-5} m is obtained, this is usually conservative as assuming flat plate friction tends to give larger u_τ than a laminar region would, however, the post run y^+ distribution was checked in order to confirm $y^+ < 1$ around the airfoil as shown in Appendix B, Figure B1(a).

Mesh refinement was prioritised at the leading and trailing edges (minimum spacing 10^{-3}), along the suction surface where separation is anticipated and in the wake region (Bashir et al., 2024). Quality metrics were maintained at min Eriksson skewness > 0.8 (>>0.5), the 2x2x2 determinant is above 0.8 (>>0.2) and angle above ~50° (>>18°) as per the ICEM CFD documentation, standards which allow for robust convergence, metrics shown in Appendix A, Figure A1 (a,b,c).

2.3.3. Flow Solver Setup

Simulations were conducted in ANSYS Fluent using a steady state, two dimensional, incompressible (freestream Mach number is low ~0.04 so compressibility is negligible), pressure based solver with the coupled pseudo transient formulation for better stability and convergence (Golmirzaee & Wood, 2024). The k-ω SST turbulence model was selected due to its superior performance in predicting adverse pressure gradients, trailing separation and near wall behaviour and typically outperforms k-ε variants in validation against experimental data (Golmirzaee & Wood, 2024). Although transition sensitive models (γ-Re_θ) were attempted, these

produced physically unstable oscillatory solutions and failed to reach convergence. Therefore, a fully turbulent k-ω SST simulation was employed, which provides numerical robustness, while acknowledging flow may be partially transitional and handle flow separation less accurately as it does not solve laminar to turbulent transition (Menter et al., 2006).

Boundary conditions included a uniform velocity inlet (3 m/s or 15 m/s with components $U_\infty \cdot \cos(10°)$, $U_\infty \cdot \sin(10°)$) for angle of attack and turbulence intensity of 5% to simulate urban environments , zero gradient pressure outlet and stationary no slip walls on the airfoil. Spatial discretisation used second order upwind for turbulence scalars with the coupled pressure velocity solver. Solution monitors tracked lift (C_l) and drag (C_d) coefficients, residuals (convergence criterion of < 10^{-6}) and mass imbalance (<10^{-6}), with convergence declared after ~200 iterations when forces stabilised within 0.1% (Golmirzaee & Wood, 2024), as shown in Appendix B, Figure B1 (b,c,d).

2.4. Mesh Independence Study

A mesh independence study was conducted at the higher wind speed of 15m/s (Re_c ~4.3×10^5) and AoA = 10°. Four progressive mesh levels were generated for independence testing the details of which are shown in Table 1, in which mesh density increases were from refinement around critical regions.

Mesh Level	Total Cells	C_l	ΔC_l (%)	C_d	ΔC_d (%)
Coarse	20256	1.294	-	0.0272	-
Medium	38016	1.306	+0.93	0.0264	-2.94
Fine	56782	1.311	+0.38	0.0262	-0.76
Extra-fine	112266	1.313	+0.15	0.0261	-0.38

***Table 1**. Mesh Independence Study*

The convergence trend shows a consistent reduction in error with refinement, C_l increases towards an asymptote while C_d decreases, with no oscillatory behaviour. Changes between the medium and fine grids are below 1% for both coefficients, indicating near grid independent forces for practical purposes, seen visually in Appendix A, Figure A1 (d).

For 2D grids, the representative grid spacing was taken as $h \propto 1/\sqrt{n}$ and discretisation uncertainty was quantified using the Grid Convergence Index (GCI) method based on Richardson extrapolation (Celik et al., 2008). GCI calculations (3 grid: fine = 1, medium = 2, coarse =3):

Refinement ratios (Celik et al., 2008, convention) are $r_{21} = \frac{h_2}{h_1} = \sqrt{N_1/N_2}$ and $r_{32} = \frac{h_3}{h_2} = \sqrt{N_2/N_3}$. For the fine/medium/coarse set (56,782/ 38,016/ 20,256 cells), $r_{21} = 1.22$ and $r_{32} = 1.37$.

Because the spatial accuracy of the second order upwind scheme is p=2 and because "apparent" orders can be inflated for small refinement ratios, GCI is reported using p = 2 with the standard safety factor F_s=1.25.

	r_{21}(fine/med)	r_{32}(fine/med)	p	GCI_{21} (fine grid, %)
C_l	1.22	1.37	2.0	0.97
C_d	1.22	1.37	2.0	1.93

***Table 2**. Grid Convergence Metrics*

The fine grid GCI is therefore ≈ 1% for C_l and ≈ 2% for C_d (Table 2), confirming discretisation uncertainty at the fine/medium resolution level. The asymptotic range condition was assessed using the GCI ratio $AR = GCI_{32}/(r_{21}^{p} GCI_{21})$ (Roache, 2009). For C_l AR≈0.91, indicating near asymptotic behaviour, however for C_d, AR≈1.5 which suggests that drag is more sensitive to mesh resolution and has not fully entered the asymptotic convergence range.

Given that the change from medium to fine is <1% for both C_l and C_d and differs less than the estimated discretisation uncertainty, the medium mesh (38,016 cells) was adopted for the remaining simulations to balance accuracy and computational costs. The final mesh is shown in Figure 1.

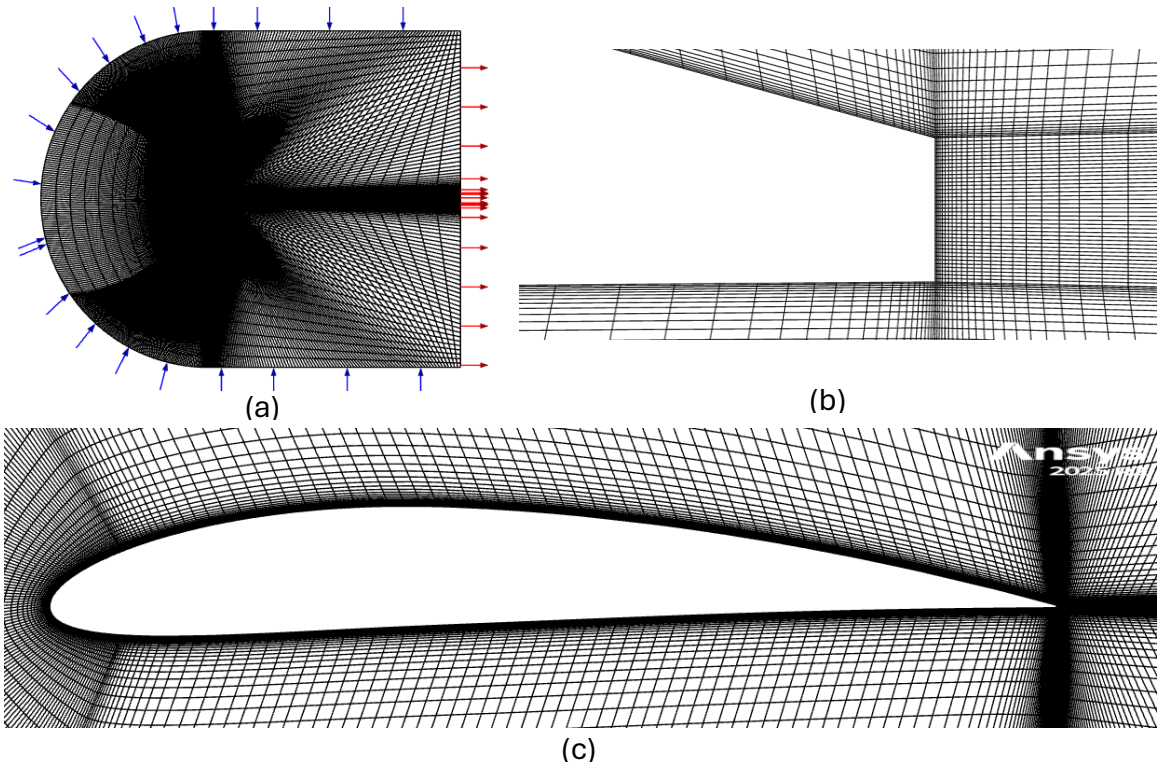


***Figure 1.** Final Mesh, (a) Full Domain, (b) Trailing Edge Transition, (c) Airfoil Close-Up*

2.5. Model Validation

To verify the CFD setup prior to the full study, the NACA 4412 airfoil was simulated at U=15m/s and AoA = 10° (medium mesh) and compared against wind tunnel measurements at a comparable Reynolds number (Re = 5x10^5). It should be noted that a turbulence intensity of 0.2% and length estimated as $\ell = \frac{0.07L}{C_{\mu}^{\frac{3}{4}}}$, where $C_{\mu} is\ 0.09$, and L is the characteristic length of the wind tunnel (Texas A&M), following an approach used by (Meana-Fernández et al., 2019) was used in these simulations to better align with the experimental data.

Experimental reference data were taken from(Ostowari et al., 1985), who report force balance measurements for NACA 44XX blades over Re = 0.25-1 x10^6 and AR = 6,9,12,∞. Their AR = ∞

configuration was selected because it is intended to represent an effectively two dimensional blade in the tunnel test section and is therefore the closest match to the present 2D CFD model.

At Re = 0.5 x 10^6 and AoA = 10°, digitisation of the published force balance plots for NACA 4412 gives $C_l \approx 1.1$ and $C_d \approx 0.055$. The present CFD predicts $C_L = 1.31$ and $C_d = 0.026$, corresponding to a +19% difference in lift and a -52% difference relative to force balance drag. The overprediction is consistent with the fully turbulent modelling assumption discussed earlier.

Ostowari and Naik also performed wake rake drag surveys for the AR=∞ blades over AoA ∈[-10°,12°] and state that the difference between force balance drag and wake rake drag represents end plate tare and tunnel interference effects associated with the tunnel floor/ceiling. Since these effects aren't present in a idealised 2D numerical model, the wake rake drag is a more appropriate reference for section profile drag in attached flow. Using their wake rake value at Re=0.5 x 10^6, AoA = 10° ($C_{d,wake} \approx 0.023$), the CFD drag is +14% higher.

The authors note that the cantilever mounting allowed bending/twisting at high Re and high lift, which can reduce lift and increase drag, aligning with the present CFD, however, no correction was applied to C_l or lift slope for this effect. For context, Ostowari and Naik report a pre stall lift curve slope of ≈ 0.09 per degree for NACA 4412, whereas (Abbott & Von Doenhoff, 1959) present section data from low turbulence tunnel tests (Re= 3-9 x10^6) and report lift curve slopes close to thin airfoil theory (~0.1 per degree) with no formal change in slope over their Re range. The lower slope in Ostowari and Naik therefore indicates that their AR = ∞ dataset may deviate from ideal rigid 2D behaviour, and is treated as a limitation of the dataset rather than a definitive explanation for the single validation point discrepancy.

For additional context only, an XFoil polar at Re=5x10^5, AoA = 10° predicts $C_l \approx 1.39$ and $C_d \approx 0.021$, placing the present CFD between experimental and XFoil predictions.

Overall, the CFD reproduces pre-stall behaviour at Re≈5x10^5 with lift agreement within ~20% and profile drag within ~14% relative to wake rake measurements at AoA=10°, providing reasonable confidence for the forthcoming study.

3. Results and Discussion

3.1. Analysis of the Baseline NACA 4412 Airfoil

This section reports key aerodynamic performance metrics (lift and drag coefficients) and use pressure, velocity/pathline and skin friction outputs to interpret the flow physics and demonstrate the simulations capture the expected behaviour, Table 3 shows the computed aerodynamic coefficients for the original airfoil at both wind speeds.

Wind Speed (m/s)	C_l	C_d	Reynolds Number	C_l/C_d
15	1.306	0.0264	4.3x10^5	49.5
3	1.154	0.0400	8.5x10^4	28.9

***Table 3**. Baseline NACA 4412 Aerodynamic Coefficients*

The airfoil produces positive lift at both wind speeds but performance degrades at the lower wind speed: C_d increases from 0.0264 to 0.0400 while the C_l decreases from 1.306 to 1.154,

because of this C_l/C_d drops from 49.5 at 15 m/s to 28.9 at 3 m/s, implying that viscous and separation related losses have a stronger impact in the lower Re case.

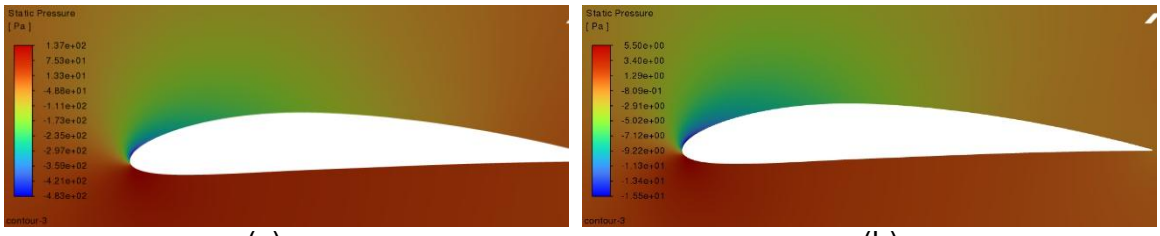
(Genç et al., 2016) shows experimental measurements (C_l and C_d) for NACA4412 at low Reynolds numbers ($2.5x10^4$ - $7.5x10^4$) and provides flow visualisations. For the 3m/s case at AoA = 10°, C_l shows a ~19% reduction in lift coefficient (Re = $7.5x10^4$ experiment), consistent with previous validation for 15 m/s, confirming general repeatability.

3.1.1. Static Pressure Contours

Figures 2 show the expected pressure distribution for a cambered airfoil at positive incidence. A higher pressure region forms near the lower leading edge where flow deaccelerates towards a stagnation point which increases pressure as kinetic energy is converted into pressure at the wall. A low pressure region (suction peak) is seen on the upper surface near the leading edge due to rapid flow acceleration around the curved suction side, downstream of this, the pressure gradually becomes higher towards the trailing edge as the flow deaccelerates and returns towards the freestream static pressure.

The net pressure difference between the lower and upper surfaces generates lift, confirming the simulation captures the fundamental lift producing physics. At 3 m/s the static pressure ranged from P_{max}≈ 5.5 Pa to P_{min}≈-15.5 Pa, with the peak at the upper leading edge, at 15 m/s the peak is located in the same however sees larger magnitudes overall (P_{max}≈ 137 Pa to P_{min}≈-483 Pa) which reflects the stronger pressure at higher wind speeds.

Pressure magnitudes should roughly scale with dynamic pressure $q = \frac{1}{2}\rho V^2$. Increasing wind speed from 3 to 15m/s (x5), should increase pressure levels by ~ 5^2=25 times, matching the observed change (≈5.5 Pa to ≈137 Pa) therefore supporting realistic boundary conditions.



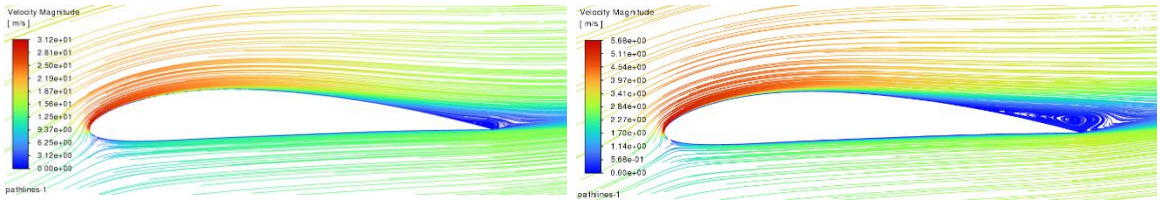
(a) (b)

***Figure 2**. Pressure Contours, (a) 15 m/s, (b) 3 m/s*

3.1.2. Velocity Magnitude, Pathlines and Separation Behaviour

Velocity magnitude and pathlines (Figures 3) support the pressure based interpretation and highlight differences in separation behaviour between the two Re regimes. At 3 m/s, the maximum velocity magnitude was ~5.68 m/s along the suction surface followed by medium then low velocities near the trailing edge. The pathlines show a clear recirculating region on the suction side indicating separated flow near the trailing edge region which contributes to the higher drag and reduced C_l/C_d at 3m/s. At 15 m/s the maximum velocity magnitude increased to ~31.2 m/s and the recirculation is seen closer to the trailing edge, meaning flow remains attached over a large portion of the suction surface. This reduction aligns with the lower drag

coefficient at 15 m/s, because delayed separation typically reduces wake thickness and pressure drag penalties.

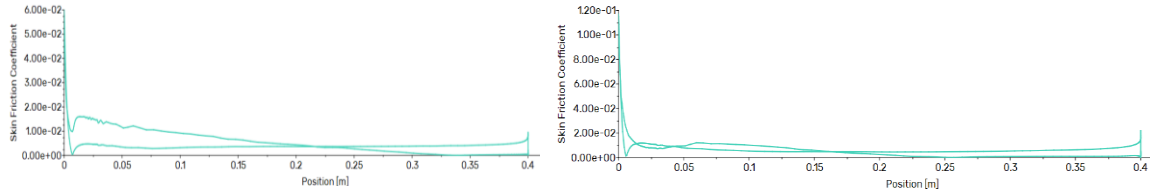


(a) (b)

***Figure 3**. Velocity Magnitude Pathlines, (a) 15 m/s, (b) 3 m/s*

The skin friction coefficient (Figure 4) gives an indicator of separation location because C_f tending to 0 corresponds to loss of wall shear stress and typically the start of a separated region. For 3m/s C_f reaches ~0 at x≈0.22m suggesting separation around x/c ≈ 0.55. This early separation aligns with the increased drag (C_d = 0.0400) and reduced efficiency (C_l/C_d=28.9) at the lower wind speed. For 15 m/s, C_f reaches ~0 further downstream at x≈0.32m (x/c≈0.8), which aligns with the lower drag coefficient (C_d = 0.0264) and higher efficiency (C_l/C_d = 49.5).

The same general trend can also be observed in (Genç et al., 2016) where the flow visualisations show separation becoming more severe as Reynolds number decreases. The authors also note laminar separation bubbles in this Re range, further confirming transitional behaviour which may not be accurately captured in fully turbulent k-ω SST, supporting the overprediction of C_l due to delayed separation and a more attached boundary layer.

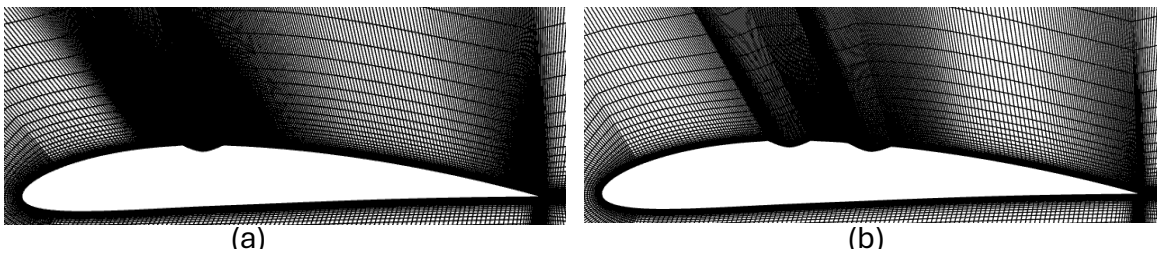


(a) (b)

***Figure 4.** Skin Friction Coefficient, (a) 15 m/s, (b) 3 m/s*

3.2. Evaluation of the Modified Airfoil Design

The modified airfoil with the suction side dimples were simulated using a similar mesh setup as validated in section 2.5, however, is more refined around the dimples as seen in (Xie et al., 2022), a close up of the suction side mesh is shown in Figure 5. This section presents the aerodynamic coefficients for both configurations and compares them with the baseline.



(a) (b)

***Figure 5**. Local Mesh Refinement, (a) Single Dimple, (b) Double Dimple*

3.2.1. Aerodynamic Performance

Table 4 shows the computed aerodynamic coefficients for the single and two dimple setups alongside the baseline for direct comparison.

Configuration	Wind Speed (m/s)	C_l	C_d	C_l/C_d	ΔC_l(%)	ΔC_d(%)
Baseline	15	1.306	0.0264	49.5	-	-
1 Dimple (x/c=0.35)	15	1.262	0.0313	40.3	-3.4	+18.6
2 Dimples (x/c=0.35,0.5)	15	1.249	0.0332	37.6	-4.4	+25.8
Baseline	3	1.154	0.0400	28.9	-	-
1 Dimple	3	1.144	0.0422	27.1	-0.9	+5.5
2 Dimples	3	1.141	0.0432	26.4	-1.1	+8.0

Table 4. *Modified Profile Aerodynamic Coefficients*

Instead of the intended objective of improving performance, both modified configurations showed higher drag and slightly reduced lift compared with the baseline at both wind speeds, with the two dimple case showing larger penalties than the single dimple configuration. At 15 m/s adding one dimple increased C_d by 18.6% and reduced aerodynamic efficiency (C_l/C_d) from 49.5 to 40.3 (-18.6%), while the two dimple configuration further degraded efficiency to 37.6 (-24% relative to baseline). At 3 m/s, the penalty is smaller in absolute terms but still negative, with single dimple increasing C_d by 5.5% and reducing efficiency by 6.2%, with the two dimple case showing similar trends.

This outcome is reasonable and aligns with known dimple control limitations documented by Xie et al, in which dimple misplacement, dimensioning or if the flow Re doesn't support the intended vortex shedding mechanism, can increase drag without providing separation reduction benefits. The differences between dimples also exceed the discretisation uncertainty meaning these results are physically meaningful.

3.2.2. Comparison with Baseline

Static pressure contours for the modified airfoils were very similar to the baseline case, with the same general features (leading edge stagnation, suction peak and pressure recovery), only very minor spikes could be seen in the dimples, however, were not large enough to affect the global pressure field.

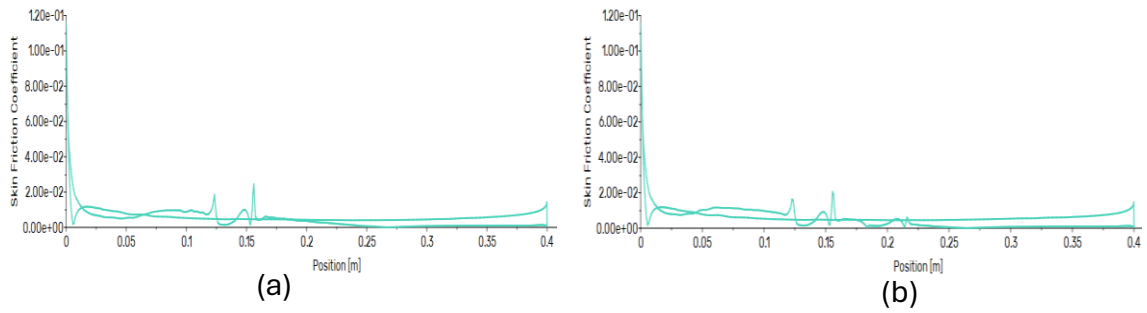


Figure 6*. Skin Friction Coefficient for Modified profile at 3 m/s, (a) one dimple, (b) 2 dimples*

The skin friction coefficient distribution (Figure 6) similarly showed the same general separation behaviour as the baseline with C_f tending to zero in approximately the same locations, which indicates the dimples did not successfully delay or reduce the separation region. Small abnormalities were seen at the dimple edges which aligns with flow disturbances at these features however, they did not translate into beneficial separation control.

Velocity magnitude and pathline plots (Figure 7) confirm that the overall separation behaviour remained largely the same with the dimples. For both cases the pathlines showed some possible recirculation or swirling motion within the dimple cavity, but this did not energise the downstream boundary layer enough to reduce the separated region.

Turbulent kinetic energy contours plots (Figure 8 for 3 m/s) also provide additional insight, the maximum TKE in the flow increased from ~0.39 to ~0.49 for both modified cases, with the peak TKE concentrated inside the dimple cavities rather than near the trailing edge. This suggests that the dimples generated local turbulence and mixing however was confined to the cavity and didn't effectively trip or reenergise the boundary layer as intended.

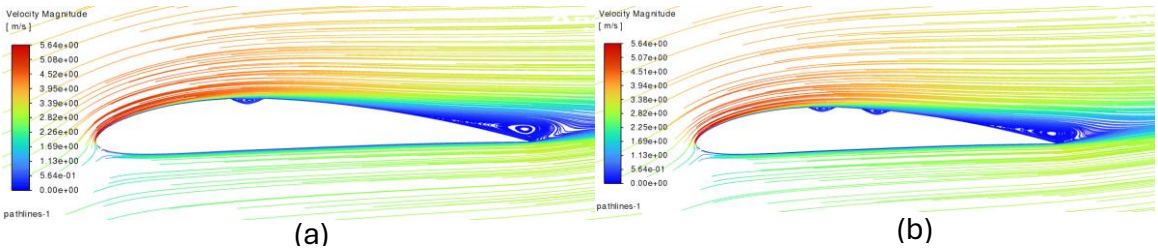


Figure 7. *Velocity Magnitude Pathlines at 3m/s, (a) One Dimple, (b) Two Dimples*

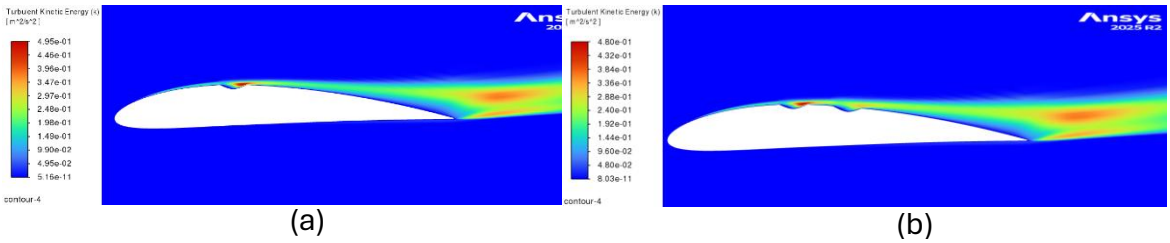


Figure 8. *Turbulent Kinetic Energy Contour, (a) One Dimple (b) Two Dimples*

3.2.3. Interpretation: Why the Dimples Did Not Improve Performance

The drag penalty and lack of separation reduction can be explained by comparing the present flow conditions with design guidelines and successful cases reported in literature.

2D modelling limitations: (Xie et al., 2022) used 3D large eddy simulations and showed that dimple effectiveness relies on three dimensional vortex structures that promote earlier transition and reattachment. In this study the 2D steady RANS simulation with fully turbulent k-ω SST can't fully capture these 3D mechanisms, the dimple acts more like an infinite groove which might add drag without generating the beneficial mixing observed by Xie et al.

At 3 m/s, the baseline separation was at around x/c ≈ 0.55, so both dimples were upstream of separation, with the second being only x/c = 0.05 away from the separation point, which is consistent with placing the dimple before separation or bubble. However, (Lan et al., 2011)

showed that location alone is not sufficient, and is dependant on whether the local boundary layer thickness relative to dimple depth supports strong near wall vorticity structures that actually energise the boundary layer. Since the contours only show local disturbances and unchanged separation, it is likely that the depth or position combination for this specific case didn't produce strong enough mixing under the simulation setup to generate any benefit.

3.2.4. Implications for VAWT Operation

VAWT blades experience a wide range of AoAs through each rotation, so separation / reattachment can occur repeatedly, so efficiency should be judged on a cycle averaged torque rather than instantaneous at a fixed AoA. In this study, at AoA = 10° the dimple seems to act primarily as disturbance even more so at higher Re when the flow is more attached. For a VAWT, this could reduce efficiency during the parts of the cycle where lift driven torque is the strongest, however if the dimple was optimised for the more separated lower Re regions, it could help reenergise the boundary layer and in turn improve average torque over the full cycle., therefore, this should be assessed in a VAWT representative test.

4. Conclusion

This work aimed to assess and improve the aerodynamic performance of a NACA 4412 blade section for urban VAWT operation using CFD at two wind speeds. It achieved this by establishing a numerically credible baseline, then demonstrating that the selected suction sided dimple did not improve performance under the tested conditions. Urban wind energy remains attractive but challenging because built environments typically have lower mean wind speeds and higher turbulence, so airfoil concepts must be robust to variable and low Re operation rather than optimised for a single condition.

A mesh independence study at 15 m/s showed smooth convergence of C_l and C_d with refinement and low discretisation uncertainty at the final selected mesh, which supports that the reported coefficients are not errors caused by mesh coarseness.

The baseline NACA4412 simulations at 3 and 15 m/s quantified the expected Reynolds number uncertainty relevant to urban VAWTs with noticeably worse aerodynamic efficiency at the lower speed consistent with increased separation losses at low Re.

A clear suction side dimple feature was modelled however, the performance objective was not met as the modified cases increased drag and slightly reduced lift when compared to the baseline at both wind speeds. The very similar global pressure fields and separation behaviour show that the dimples added losses without giving effective boundary layer energisation, which in turn would reduce the torque generation of a VAWT operating under similar conditions.

The negative result however, is still practically valuable as it indicates that passive flow control concepts are highly sensitive to sensitive to Reynolds number, boundary layer state and suitable design parameters. In built up environments where wind is intermittent and turbulence is elevated device placement and geometry would require case specific optimisation and validation with higher fidelity modelling (e.g. URANS/LES) and experiments before transferring a concept into a VAWT blade design.

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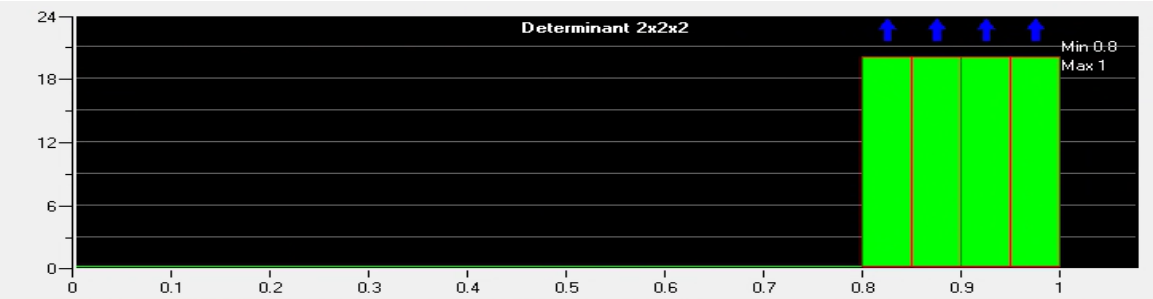
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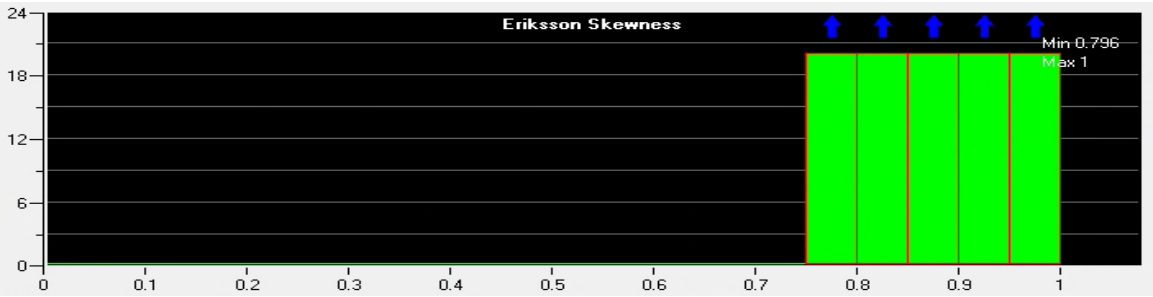
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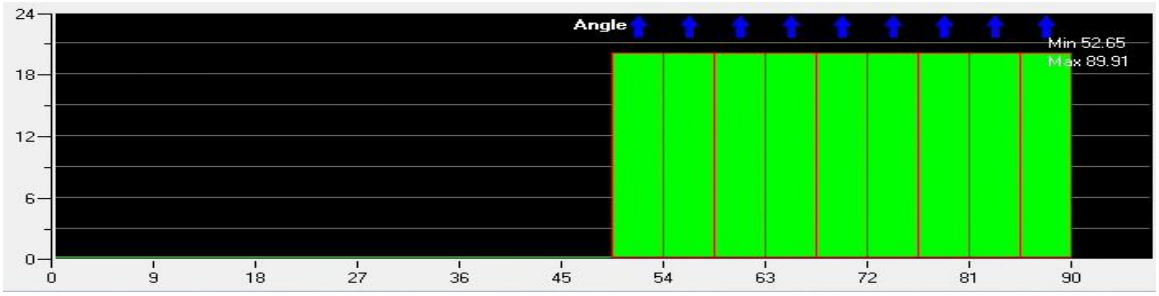
Appendix A



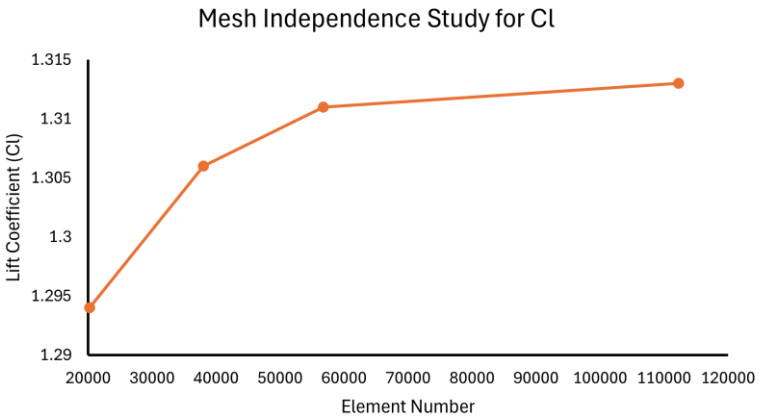
(a)



(b)



(c)



(d)

Figure A1. Mesh metrics and Independence, (a) 2x2x2 Determinant, (b) Eriksson Skewness, (c) Angle, (d) Mesh independence for Cl

(a,b,c) shows the minimum determinant of 0.8, minimum Erikson skewness of ~0.8 and minimum angle of ~53 degrees. (d) Shows the convergence trend for the lift coefficient.

Appendix B

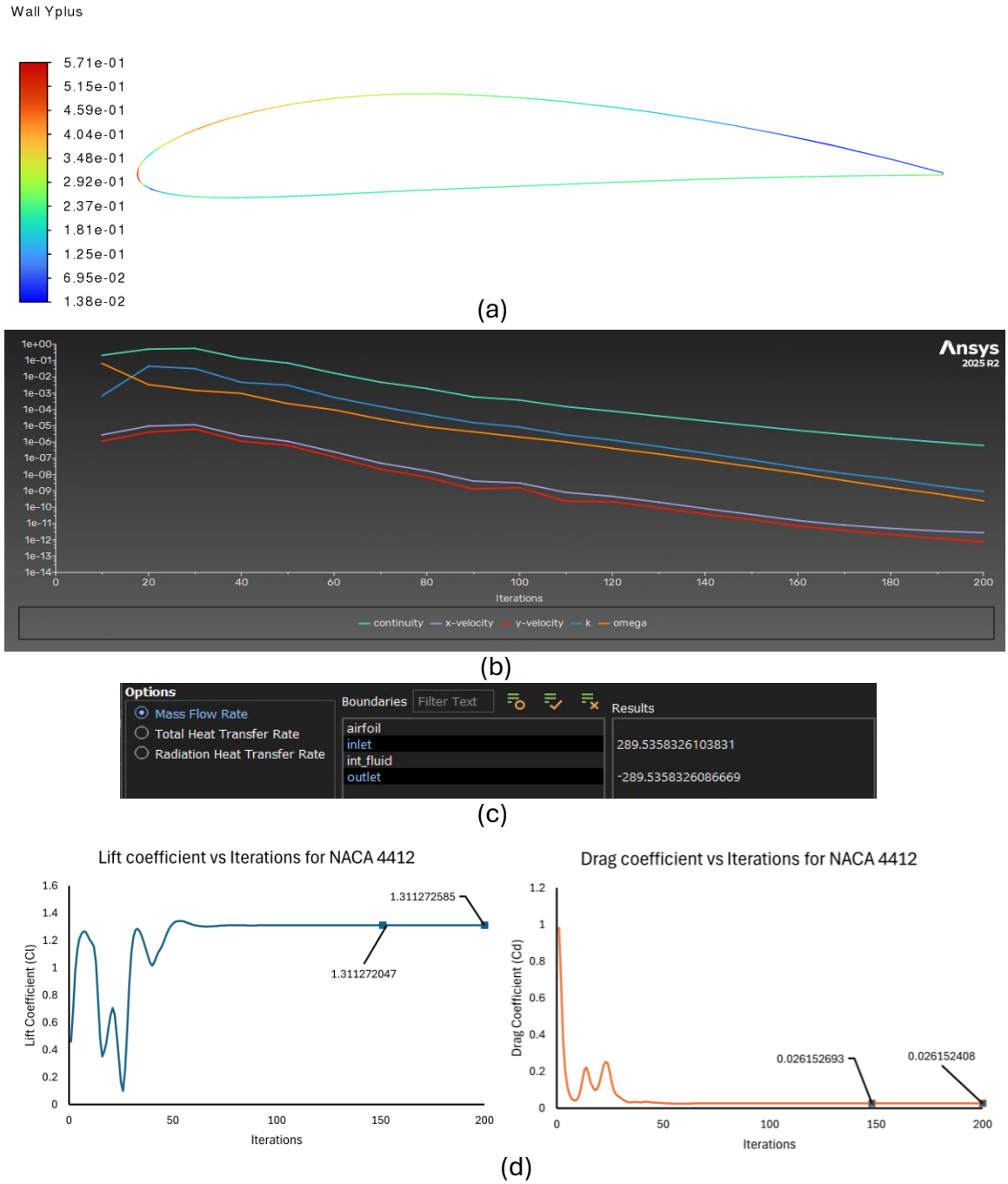


Figure B1. Convergence and y+, (a) y^+ around airfoil wall, (b) residual convergence, (c) mass fluxes, (d) force stabilisation for C_l and C_d

(a) Shows y^+ is consistently < 0.5 around the profile, with max of 0.57 at the leading edge. (b) Shows the residuals smoothly converging to $1x10^{-6}$ after ~200 iterations. (c) Shows the max fluxes at the inlet and outlet, resulting in mass imbalance in the order of 10^{-11}. (d) Shows C_l and C_d stabilising within 0.1% for the final 50 iterations.